

Problem-Solution Fit

Date	26 June 2026
Team ID	6a3a5217f0520a2a6b0309d8
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] - Banks and financial institutions - Credit analysts and loan officers - Customers applying for credit cards	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited financial information - Incomplete application details - Varying income sources - Limited credit history - Dependence on manual processing	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual Verification: Accurate but slow and costly - Traditional Rule-Based Systems: Fast but lacks adaptability - Machine Learning Models: Fast, scalable, accurate and learns from historical data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual verification is slow and time-consuming - Human errors during evaluation - Large number of applications received daily - Delays and inconsistent approval decisions	9. PROBLEM ROOT / CAUSE [RC] - Increasing application volume - Manual verification process - Inconsistent human decision-making - Incomplete applicant data - Inefficient risk assessment	7. BEHAVIOR + ITS INTENSITY [BE] - Applicants expect quick approval decisions - Banks process thousands of applications daily - Need for high-speed, consistent and reliable evaluation
3. TRIGGERS TO ACT [TR] - Increase in credit card applications - Need for faster approval - Customer demand for instant decisions - Requirement to reduce operational costs	10. YOUR SOLUTION [SL] Develop a Credit Card Approval Prediction System using Machine Learning (Logistic Regression, Decision Tree, Random Forest and XGBoost) integrated with a Flask web application. The system predicts approval or rejection instantly based on applicant information, improving accuracy, reducing processing time and assisting banks in making reliable decisions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Bank websites - Mobile banking applications - Online credit card portals OFFLINE - Bank branches - Customer service centers - Financial consultants
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration, uncertainty, long waiting time and stress After: Faster approval, confidence, transparency, improved customer satisfaction and trust		